Exam Date & Time: 02-Dec-2023 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOV 2023

OBJECT ORIENTED PROGRAMMING [DSE 2123]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions. Section Duration: 180 mins

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1) Discuss the importance of multi-level inheritance for the following scenario:
Consider *Vehicle* as the base class and its derived classes: *Car* and *SportsCar*. Define the constructors of classes as *Vehicle(String Name), Car(String name, int passenger)* and *SportsCar(String name, int passenger, Boolean turbo).* Consider start() and displayInfo() as the two overriding methods in all the classes and show how the program behaves when creating instances of Car and SportsCar and invoking their methods.

A)

Sample Output:

```
Starting the Generic Vehicle vehicle.
This is a generic vehicle: Generic Vehicle

Starting the 4-passenger car by Toyota.
This is a car named Toyota
Passengers: 4
Toyota is being driven.

Starting the 2-passenger car by Ferrari.
This is a car named Ferrari
Passengers: 2
Ferrari is a sports car with turbo and is driven at high speed.
```

(5)

B) What are the purposes of the *super* keyword in inheritance? Demonstrate the same with a suitable example. (3)

C) Develop a TrafficLight program using Java enums to represent the traffic light signals as RED, YELLOW, GREEN. (2)

2) You are asked to design an object-oriented Java program for a text processing tool by considering the following:

A) Let the TextProcessor class have convertToUpper(), convertToLower() method that converts the text to upper and lower respectively. Also, consider one more method removeDigits() that removes the digits when an alphanumerical input is given and displays only string value from it. When an input length is less than 5 characters, raise a custom exception as "StringTooShortException". Implement all the above methods in a main class by creating appropriate objects. (5)

B)	How can you access the members (fields and methods) of an *inner class* from outside the *outer class*?	(3)
C)	Explain the use of *Date* class to format and display the current date and time to the user.	(2)
3)		
A)	Explain how inheritance works with *abstract classes* in Java. Describe the relationship between an *abstract class* and its concrete subclasses. Write a program with an *abstract class* representing a geometric shape, such as "Shape," and concrete classes like "Circle" and "Rectangle" that extend the *abstract class*. Implement common methods in the *abstract class* and specific methods in the concrete classes.	(5)
B)	Write a multithreaded program in java using *Runnable interface* to perform the following operations: Main thread: Read the email ids of 'n' students from the keyboard and store them in an array of strings. Thread-1: Display all email ids ending with manipal.edu Thread-2: Display all email ids ending with gmail.com	(3)
C)	With the help of an example, illustrate the usage of *toString()* to display content of any object.	(2)
4)		
A)	Create a class named "CarInfo" that stores information about a car, such as: its unique identifier (integer), model name (string), manufacturer (string), year of manufacture (integer), and price (float). Implement the class with appropriate input and output methods. Use *serialization* to store the details of 'n' cars in a file called "Car_Details.txt". Read the information from the file "Car_Details.txt" and calculate the total cost of all the cars.	(5)
B)	Write a program to create a random-access file *numbers.txt* and store 7 double values. Perform the following operations: i.Display total size of this file (in bytes). ii.Add 15.5 to the 4th data item in the file. iii.Append a value: 22.5 at the end of the file. Write a complete program to implement the above tasks.	(3)
C)	Consider a situation where you need to read data from a file, and you want to ensure that the file is closed correctly, regardless of whether an exception is thrown or not. How do you handle it using *exception handling* mechanism?	(2)
5)		
A)	Create an *ArrayList* of strings and add 'n' strings input by the user. Process the *ArrayList* to store palindrome strings in another *ArrayList*. Display the contents of the second ArrayList using *ListIterator*	(5)
B)	Consider a scenario where you want to simulate deposit and withdrawal transactions on the same bank account in a multi-threaded Java program. Explain how you can achieve this using the join() operation to ensure that all transactions are completed before calculating the final account balance.	(3)
C)	Consider a scenario you want to create a Java program to add a *label* and a *text field* to a *frame*. Write the Java statements to achieve this task.	(2)

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